Cloud Computing

Practical No. 2: Virtualization

Aim: Installation and Configuration of virtualization using KVM.

Step 1: Install the System

```
acer@ubantu:~$ sudo apt-get install
[sudo] password for acer:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
0 upgraded, 0 newly installed, 0 to remove and 11 not upgraded.
```

Step 2: Check CPU Virtualization Support

Verify whether the processor supports virtualization:

```
acer@ubantu:~$ cat /proc/cpuinfo
processor       : 0
vendor_id       : AuthenticAMD
cpu family      : 25
model           : 80
model name      : AMD Ryzen 7 5700G with Radeon Graphics
stepping        : 0
microcode       : 0xffffffff
cpu MHz         : 3792.782
cache size      : 512 KB
physical id     : 0
siblings        : 8
core id         : 0
cpu cores       : 8
apicid          : 0
initial apicid  : 0
fpu             : yes
fpu_exception   : yes
cpuid level     : 13
wp              : yes
flags           : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov
pat pse36 clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt rdtscp lm cons
tant_tsc rep_good nopl nonstop_tsc cpuid extd_apicid tsc_known_freq pni pclmulqd
q ssse3 cx16 sse4_1 sse4_2 movbe popcnt aes rdrand hypervisor lahf_lm cmp_legacy
```

Step 3: Install KVM and Required Packages

Install essential virtualization packages:

```
acer@ubantu:~$ sudo apt-get install qemu-kvm libvirt-bin bridge-utils virt-manag
er
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'qemu-system-x86' instead of 'qemu-kvm'
Package libvirt-bin is not available, but is referred to by another package.
This may mean that the package is missing, has been obsoleted, or
is only available from another source
```

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Step 4: Perform the following commands

```
acer@ubantu:~$ sudo adduser rait
info: Adding user `rait' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `rait' (1002) ...
info: Adding new user `rait' (1002) with group `rait (1002)' ...
info: Creating home directory `/home/rait' ...
info: Copying files from `/etc/skel' ...
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
Changing the user information for rait
Enter the new value, or press ENTER for the default
        Full Name []: mac
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
info: Adding new user `rait' to supplemental / extra groups `users' ...
info: Adding user `rait' to group `users' ...
```

```
acer@ubantu:~$ sudo deluser rait
info: Removing crontab ...
info: Removing user `rait' ...
acer@ubantu:~$
```

```
acer@ubantu:~$ sudo apt install libvirt-client
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package libvirt-client
```

Step 5: Install and Launch Virtual Machine Manager

```
acer@ubantu:~$ sudo apt install virt-manager
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  cpu-checker dmeventd gir1.2-ayatanaappindicator3-0.1 gir1.2-gtk-vnc-2.0
  gir1.2-gtksource-4 gir1.2-libosinfo-1.0 gir1.2-libvirt-glib-1.0
  gir1.2-spiceclientglib-2.0 gir1.2-spiceclientgtk-3.0 i965-va-driver
  intel-media-va-driver ipxe-qemu ipxe-qemu-256k-compat-efi-roms libaio1t64
  libboost-iostreams1.83.0 libboost-thread1.83.0 libburn4t64 libcacard0
  libdaxctl1 libdevmapper-event1.02.1 libegl-mesa0 libfdt1 libgbm1
  libgl1-mesa-dri libglapi-mesa libglx-mesa0 libgtk-vnc-2.0-0
  libgtksourceview-4-0 libgtksourceview-4-common libgvnc-1.0-0 libigdgmm12
  libiscsi7 libisoburn1t64 libisofs6t64 libjack-jackd2-0 liblvm2cmd2.03
  libndctl6 libnss-mymachines libosinfo-1.0-0 libosinfo-l10n libphodav-3.0-0
  libphodav-3.0-common libpmem1 libpmemobj1 librados2 librbd1 librdmacm1t64
  libsdl2-2.0-0 libslirp0 libspice-client-glib-2.0-8 libspice-client-gtk-3.0-5
  libspice-server1 libtpms0 liburing2 libusbredirhost1t64
  libusbredirparser1t64 libva-x11-2 libva2 libvirglrenderer1 libvirt-clients
  libvirt-daemon libvirt-daemon-config-network libvirt-daemon-config-nwfilter
  libvirt-daemon-driver-qemu libvirt-daemon-system
  libvirt-daemon-system-systemd libvirt-glib-1.0-0 libvirt-glib-1.0-data
  libvirt-l10n libvirt0 libxatracker2 libxml2-utils libyajl2 lvm2 mdevctl
  mesa-va-drivers mesa-vulkan-drivers msr-tools osinfo-db ovmf
```

```
acer@ubantu:~$ sudo virt-manager
acer@ubantu:~$
```

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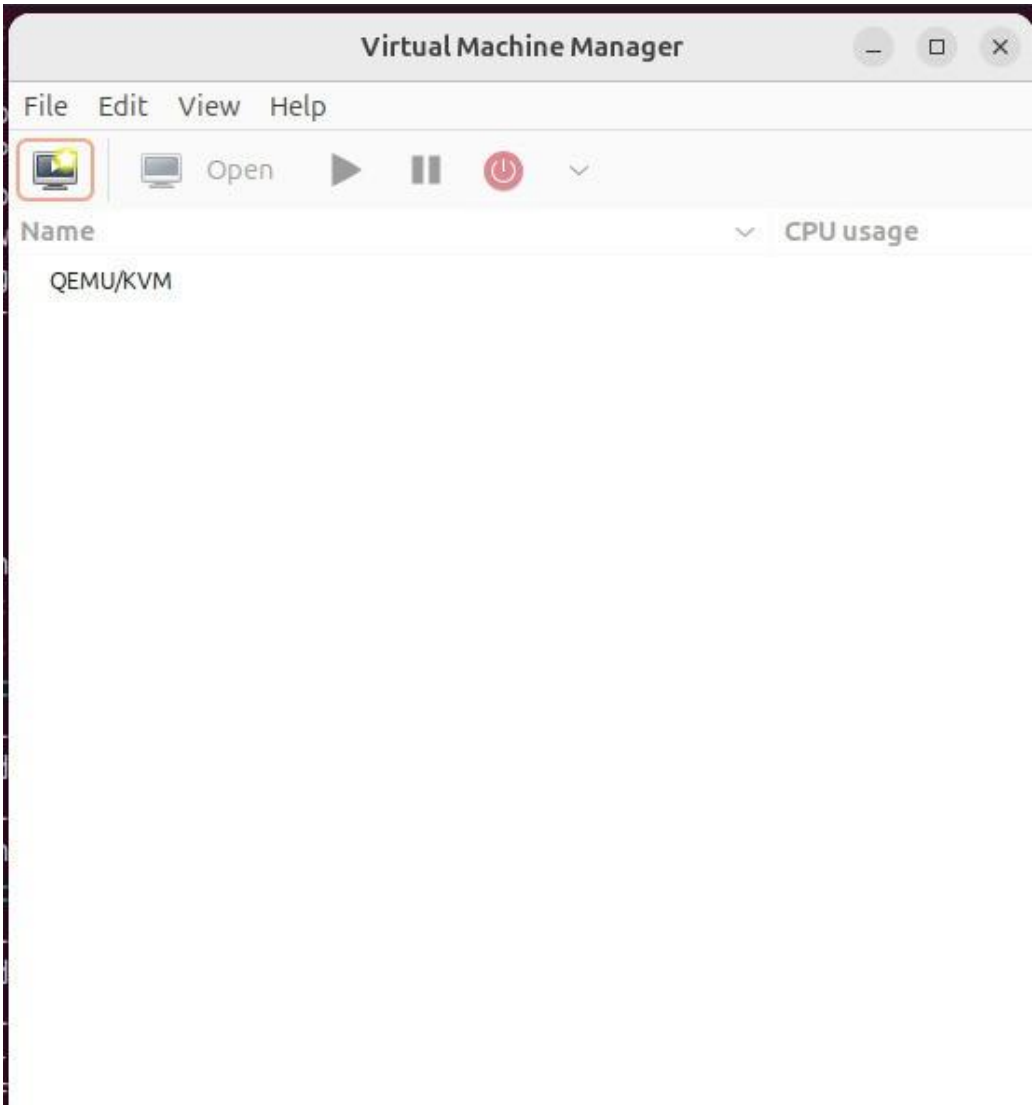
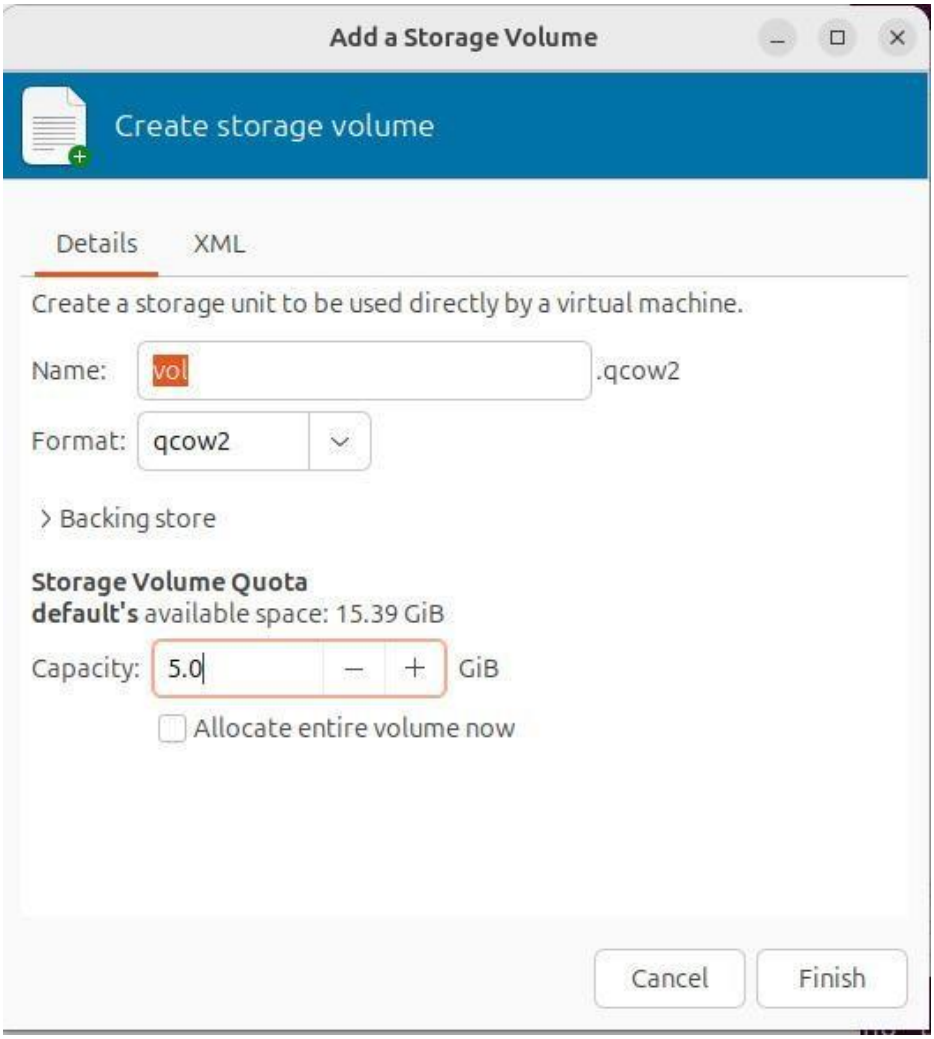
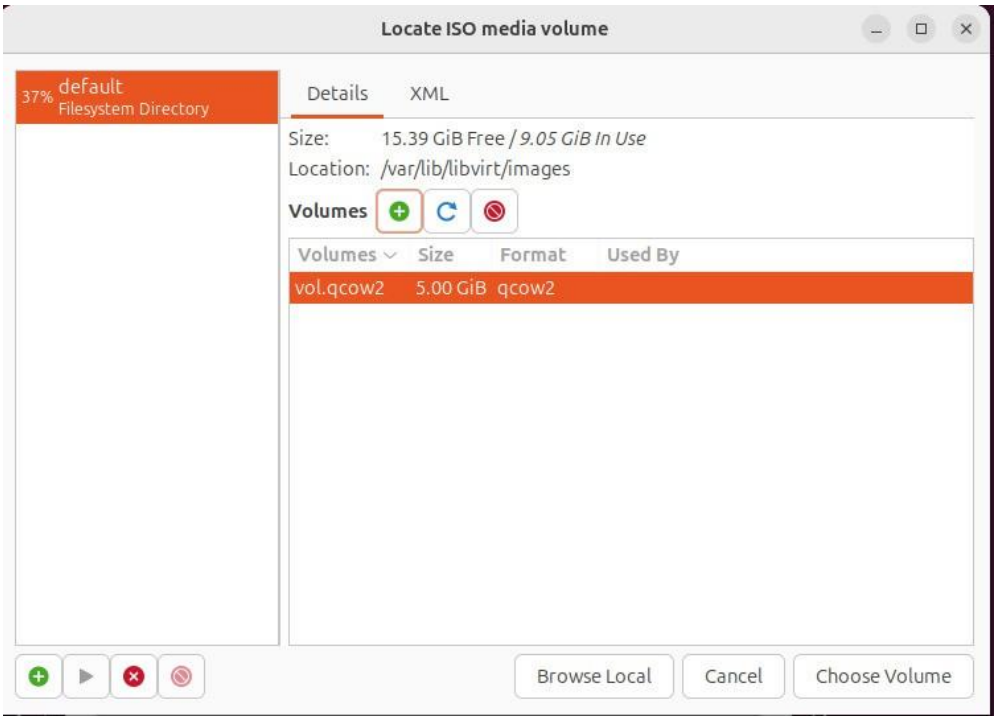
Output:


Virtual Machine Manager
File
Edit
View
Help
Open
Name
CPU usage
QEMU/KVM

Step 6: Further Follow the given following steps





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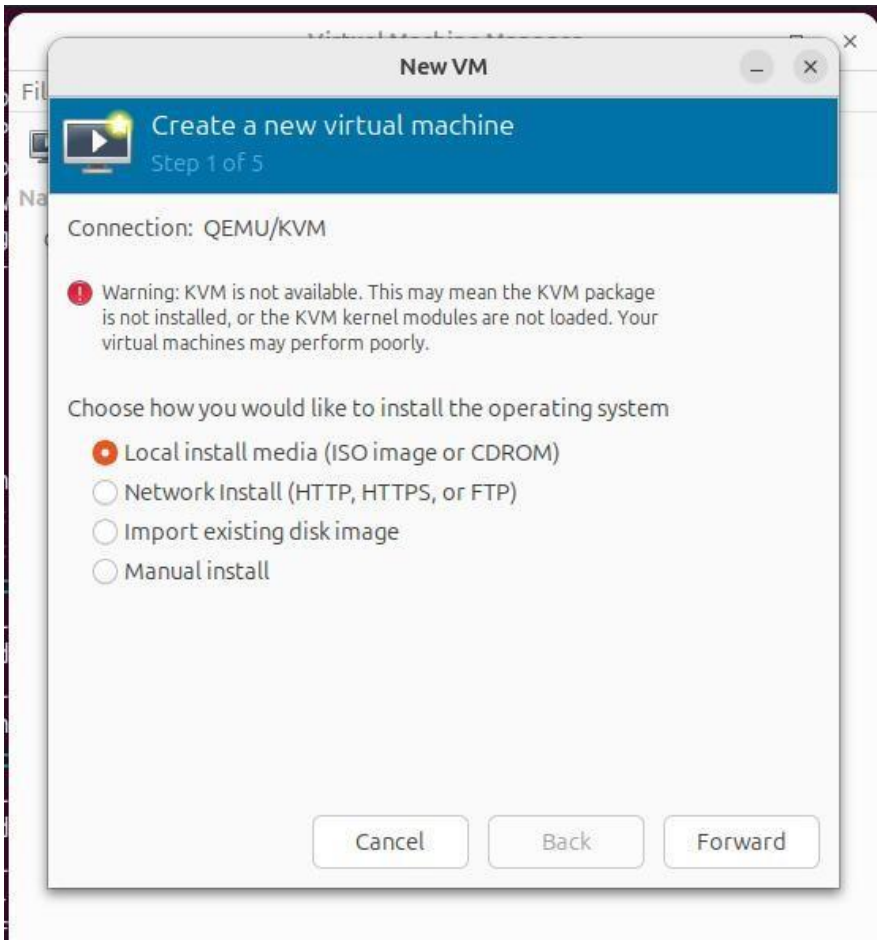
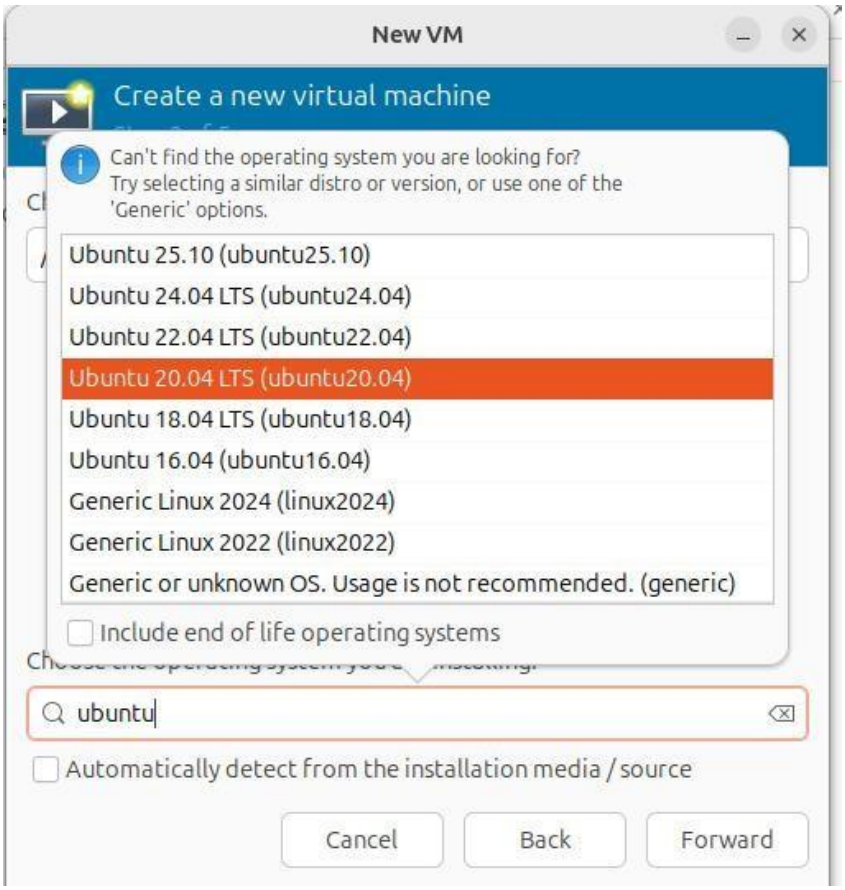
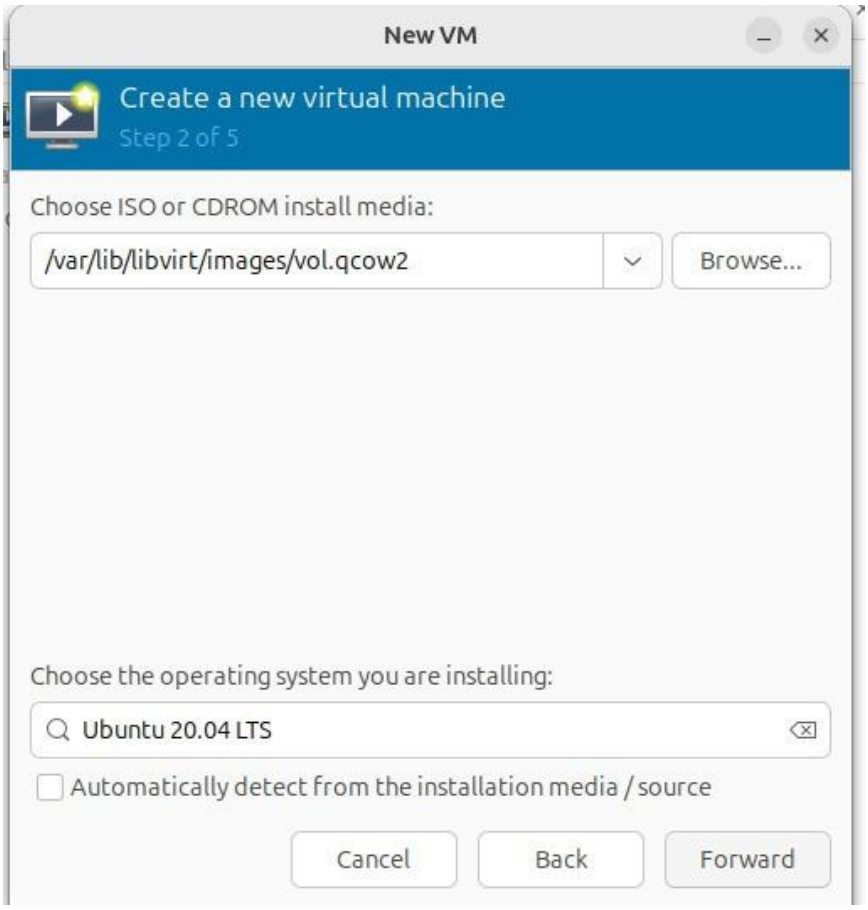
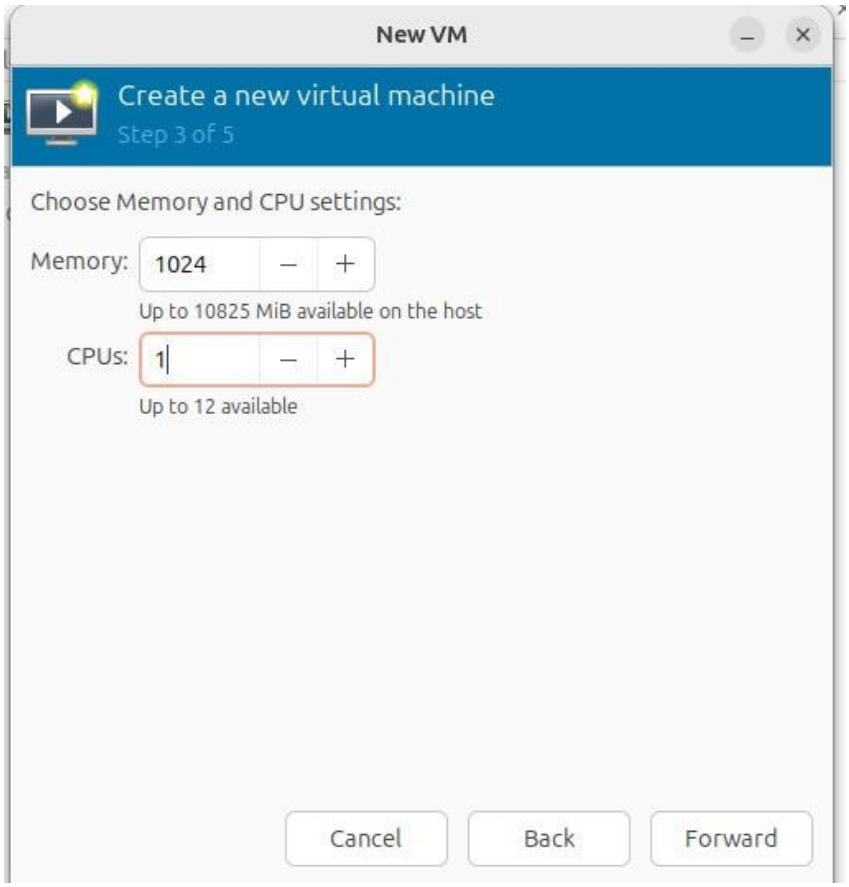
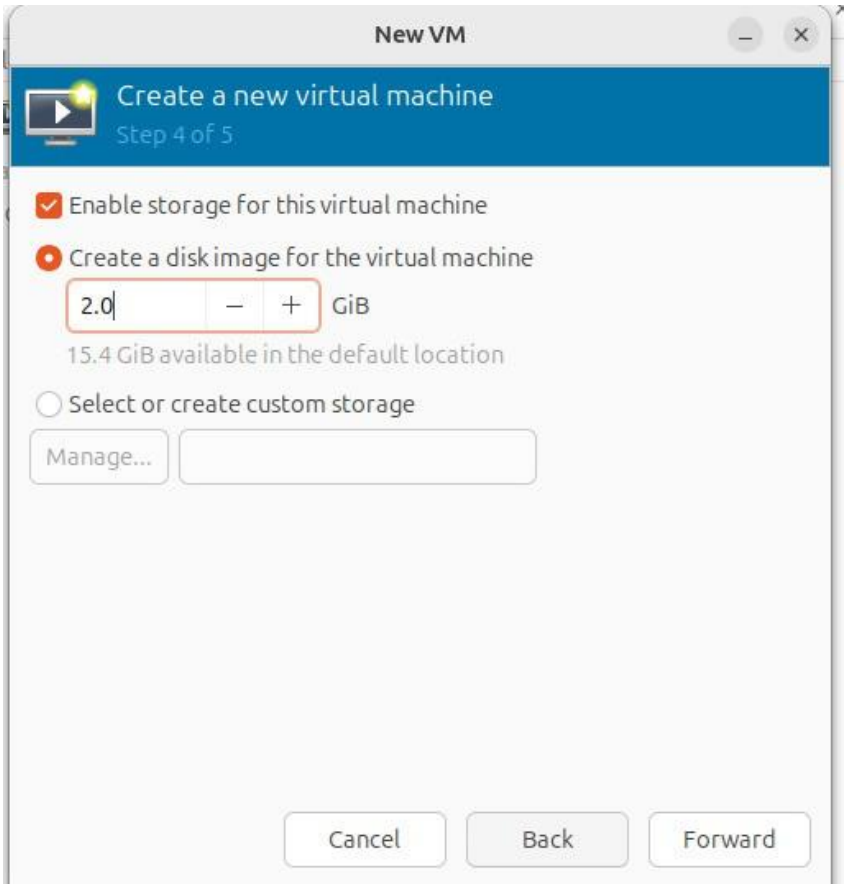
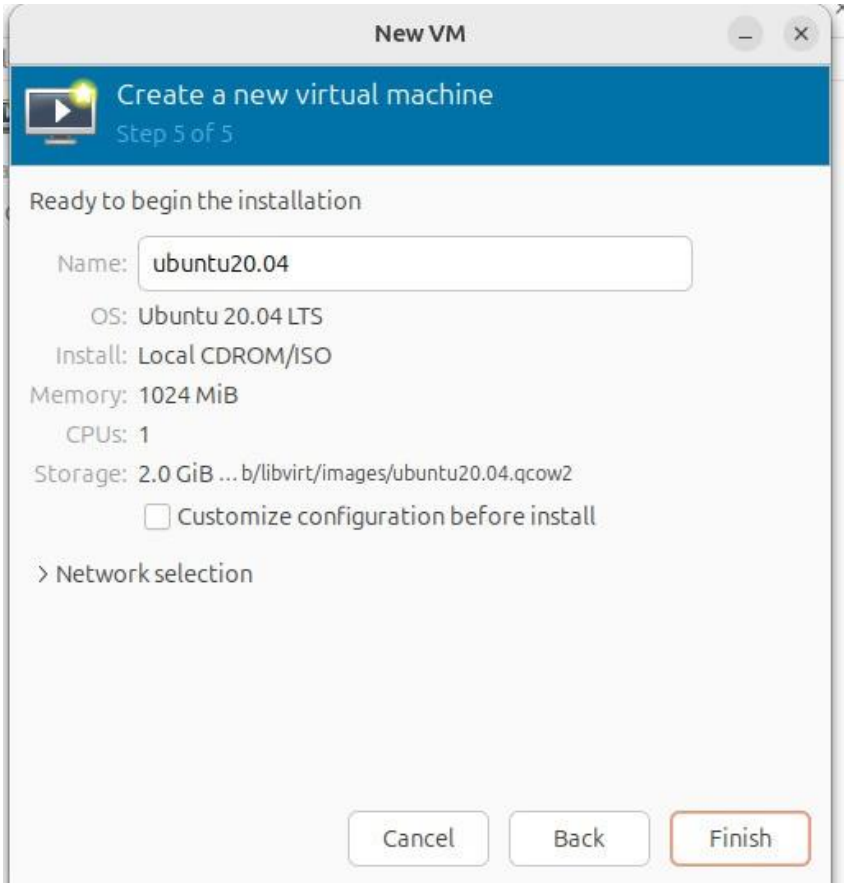

New VM
Create a new virtual machine
Step 1 of 5
Connection: QEMU/KVM
Warning: KVM is not available. This may mean the KVM package is not installed, or the KVM kernel modules are not loaded. Your virtual machines may perform poorly.
Choose how you would like to install the operating system
Local install media (ISO image or CDROM)
Network Install (HTTP, HTTPS, or FTP)
Import existing disk image
Manual install
Cancel
Back
Forward


New VM
Create a new virtual machine
Can't find the operating system you are looking for?
Try selecting a similar distro or version, or use one of the 'Generic' options.
Ubuntu 25.10 (ubuntu25.10)
Ubuntu 24.04 LTS (ubuntu24.04)
Ubuntu 22.04 LTS (ubuntu22.04)
Ubuntu 20.04 LTS (ubuntu20.04)
Ubuntu 18.04 LTS (ubuntu18.04)
Ubuntu 16.04 (ubuntu16.04)
Generic Linux 2024 (linux2024)
Generic Linux 2022 (linux2022)
Generic or unknown OS. Usage is not recommended. (generic)
Include end of life operating systems
ubuntu
Automatically detect from the installation media / source
Cancel
Back
Forward


New VM
Create a new virtual machine
Step 2 of 5
Choose ISO or CDROM install media:
/var/lib/libvirt/images/vol.qcow2
Browse...
Choose the operating system you are installing:
Ubuntu 20.04 LTS
Automatically detect from the installation media / source
Cancel
Back
Forward


New VM
Create a new virtual machine
Step 3 of 5
Choose Memory and CPU settings:
Memory:
1024
Up to 10825 MiB available on the host
CPUs:
1
Up to 12 available
Cancel
Back
Forward


New VM
Create a new virtual machine
Step 4 of 5
Enable storage for this virtual machine
Create a disk image for the virtual machine
2.0
GiB
15.4 GiB available in the default location
Select or create custom storage
Manage...
Cancel
Back
Forward


New VM
Create a new virtual machine
Step 5 of 5
Ready to begin the installation
Name: ubuntu20.04
OS: Ubuntu 20.04 LTS
Install: Local CDROM/ISO
Memory: 1024 MiB
CPUs: 1
Storage: 2.0 GiB ... b/libvirt/images/ubuntu20.04.qcow2
Customize configuration before install
Network selection
Cancel
Back
Finish